CS60 Database Concepts & Applications Final Exam Total:- 70 Points (converted to 150 Points)

Name:- Siarhei Hancharou **Id#:-1712666**

Part A:- Multiple Choice [15 Points]

Circle the letter of the best choice.

1. In the context of the database design process, the conceptual design step that determines end-user views, outputs, and transaction-processing requirements is __**a**___.
   a) data analysis and requirements
   b) entity relationship modeling and normalization
   c) data model verification
   d) distributed database design

2. The maintenance and evolution phase of the Database Life Cycle (DBLC) involves __**b**___.
   a) Defining objectives
   b) introducing changes
   c) testing the database
   d) installing the DBMS

3. At the implementation level, the supertype and its subtype(s) depicted in a specialization hierarchy maintain a(n) __**b**___ relationship.
   a) self-referencing
   b) 1:1
   c) 1:M
   d) M:N

4. A table that has all key attributes defined, has no repeating groups, and all its attributes are dependent on the primary key is said to be in __**b**___.
   a) 1NF
   b) 2NF
   c) 3NF
   d) 4NF

5. Which query is used to list a unique value for V_CODE, where the list will produce only a list of those values that are different from one another?

   SELECT ONLY V_CODE
   FROM PRODUCT;

   SELECT UNIQUE V_CODE
   FROM PRODUCT;

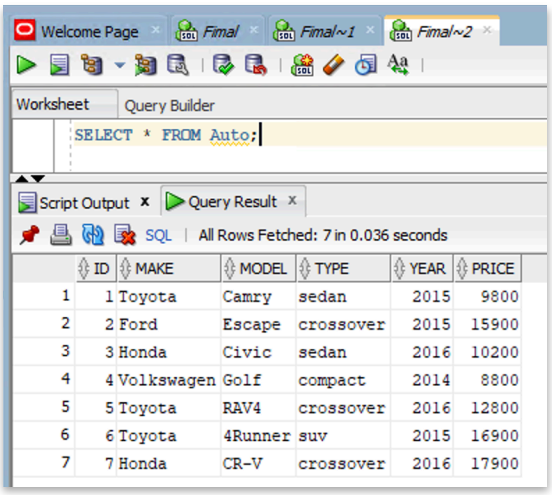
   c) SELECT DIFFERENT V_CODE
   FROM PRODUCT;

   d)
   ```
   SELECT DISTINCT V_CODE
   FROM PRODUCT;
   ```

Use the Auto Table to answer questions **6 to 10**.

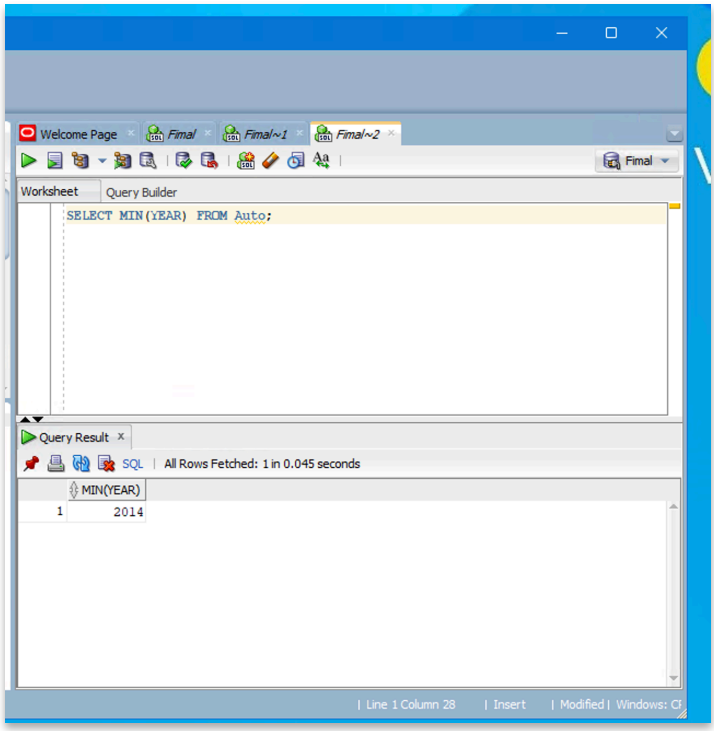
Auto

ID	Make	Model	Type	Year	Price
1	Toyota	Camry	sedan	2015	9800
2	Ford	Escape	crossover	2015	15900
3	Honda	Civic	sedan	2016	10200
4	Volkswagen	Golf	compact	2014	8800
5	Toyota	RAV4	crossover	2016	12800
6	Toyota	4Runner	suv	2015	16900
7	Honda	CR-V	crossover	2016	17900



6. Choose the correct **SELECT statement** that returns the given result:- **2014**

a) SELECT MAX(YEAR) FROM Auto;
b) SELECT MIN(PRICE) FROM Auto;
c) SELECT MIN(YEAR) FROM Auto;
d) SELECT AVG(PRICE) FROM Auto;



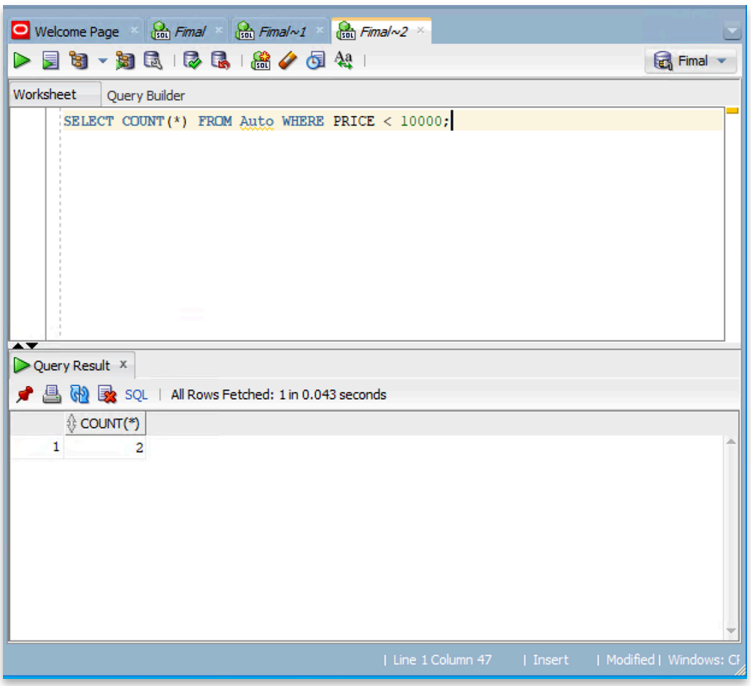
7. Choose the correct **SELECT statement** that returns the given result:- **2**

a) SELECT COUNT(*) FROM Auto;

b) SELECT COUNT(*) FROM Auto WHERE PRICE > 10000;

c) SELECT COUNT(*) FROM Auto WHERE PRICE < 10000;

d) SELECT COUNT(*) FROM Auto WHERE PRICE BETWEEN 5000 AND 12000;



8. Choose the correct **SELECT statement** that returns the given result from the AUTO Table:-

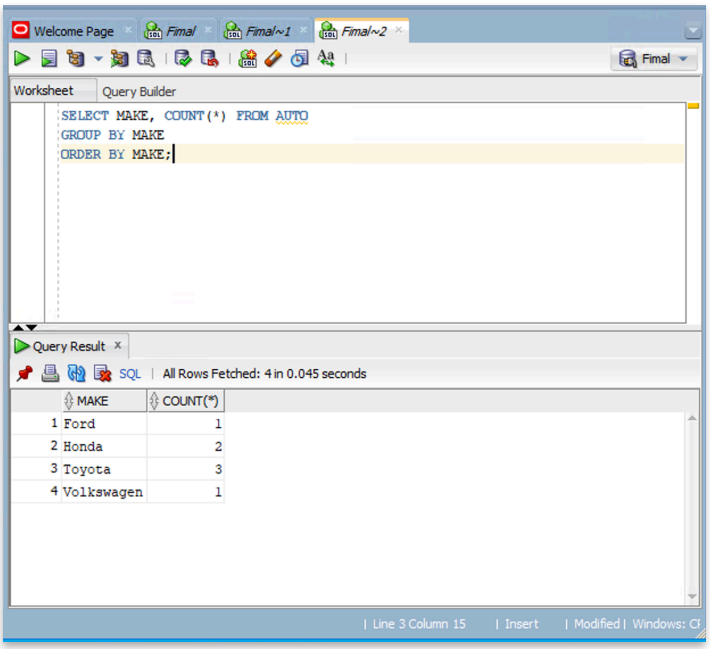
Ford	1
Honda	2
Toyota	3
Volkswagen	1

a) SELECT MAKE, COUNT(*) FROM Auto;

b) SELECT MAKE, PRICE, COUNT(*) FROM AUTO
GROUP BY MAKE
ORDER BY MAKE;

c) SELECT MAKE, COUNT(*) FROM AUTO
GROUP BY MAKE
ORDER BY MAKE;

d) SELECT COUNT(*) FROM AUTO
GROUP BY MAKE
ORDER BY MAKE;



9. Choose the correct **SELECT statement** that returns the given result from the AUTO Table:-

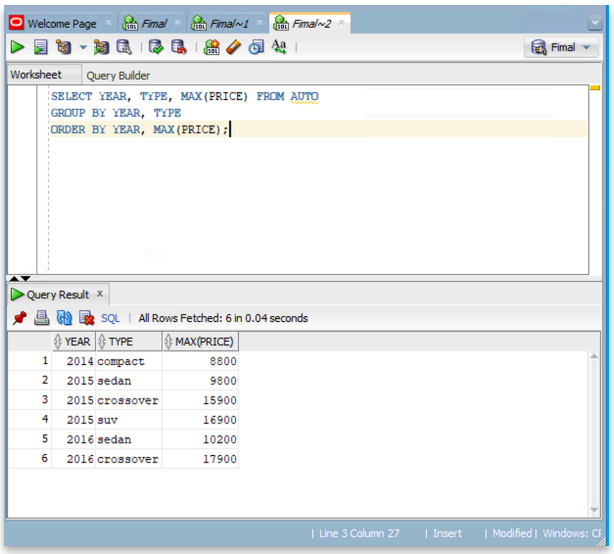
```
2014  compact    8800
2015  crossover  15900
2015  suv        16900
2016  sedan      10200
2016  crossover  17900
```

a) SELECT YEAR, TYPE, MAX(PRICE) FROM AUTO
GROUP BY YEAR, TYPE, MAX(PRICE)
ORDER BY YEAR;

b) SELECT YEAR, TYPE, MAX(PRICE) FROM AUTO
GROUP BY YEAR, TYPE
ORDER BY MAX(PRICE);

c) SELECT YEAR, TYPE, MAX(PRICE) FROM AUTO
GROUP BY YEAR
ORDER BY YEAR, MAX(PRICE);

d) SELECT YEAR, TYPE, MAX(PRICE) FROM AUTO
GROUP BY YEAR, TYPE
ORDER BY YEAR, MAX(PRICE);



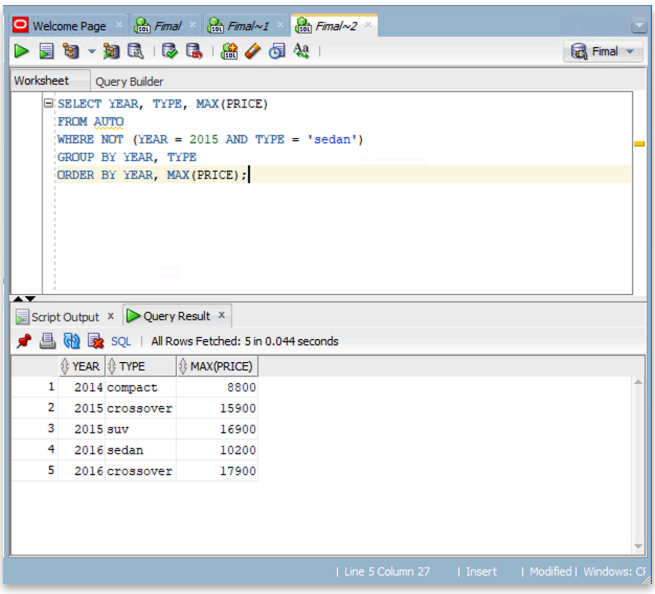
Note:

The correct SQL statement for Question 9 is **#d**

This query is logically correct and produces the full grouped result.

However, to match exactly the shortened output shown in the exam (which omits the row 2015 sedan 9800), you may use:

```
SELECT YEAR, TYPE, MAX(PRICE)
FROM AUTO
WHERE NOT (YEAR = 2015 AND TYPE = 'sedan')
```



GROUP BY YEAR, TYPE
ORDER BY YEAR, MAX(PRICE);

10. Choose the correct **SELECT statement** that returns the given result from the AUTO Table:-

```
2015  crossover  15900
2015  suv        16900
2016  crossover  17900
```

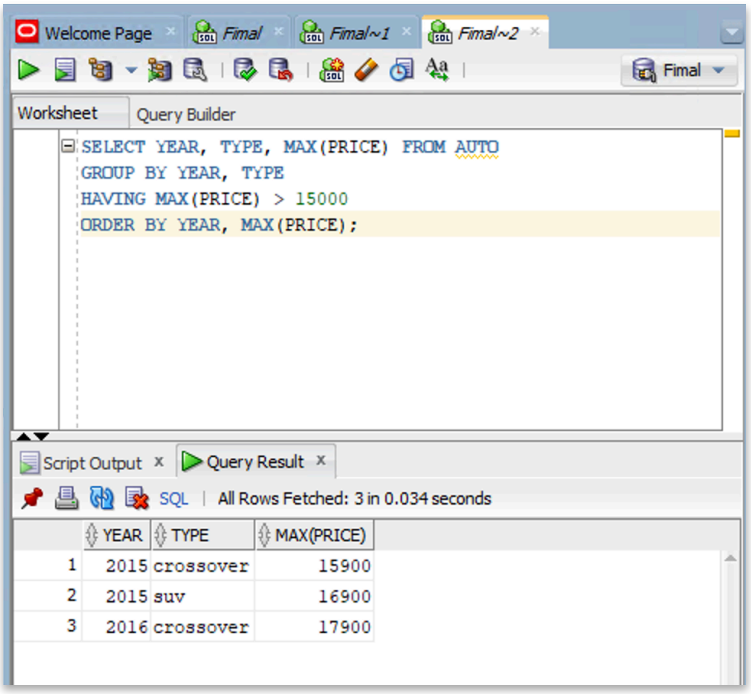
a) SELECT YEAR, TYPE, MAX(PRICE) FROM AUTO
GROUP BY YEAR
ORDER BY YEAR, MAX(PRICE);
HAVING MAX(PRICE) > 15000;

b) SELECT YEAR, TYPE, PRICE FROM AUTO
GROUP BY YEAR, TYPE, MAX(PRICE)
ORDER BY YEAR;

c) SELECT YEAR, TYPE, MAX(PRICE) FROM AUTO
GROUP BY YEAR, TYPE
HAVING MAX(PRICE) > 15000;
ORDER BY YEAR, MAX(PRICE);

d) SELECT YEAR, TYPE, MAX(PRICE) FROM AUTO
GROUP BY YEAR, TYPE
ORDER BY YEAR, MAX(PRICE);

Note: The answer is correct, but the semicolon after HAVING MAX(PRICE) > 15000 must be removed because it prematurely ends the SQL statement.



11. The syntax for a left outer join is __**b**___.
    a) SELECT column-list
       FROM table1 OUTER JOIN table2 LEFT
       WHERE join-condition
    b) SELECT column-list
       FROM table1 LEFT [OUTER] JOIN table2
       ON join-condition
    c) SELECT column-list
       WHERE LEFT table1 = table
    d) SELECT column-list
       FROM table1 LEFT table2 [JOIN]
       WHERE join-condition

12. How many rows would be returned from a cross join of tables A and B, if A contains 3 rows and B contains 5?
    a) 3
    b) 8
    c) 15
    d) 30

Refer to the **Department table** to answer **questions 13-15**.

Department

•Code	Name	ManagerID
44	Engineering	2538
82	Sales	6381
12	Marketing	6381
99	Technical support	NULL

13. Which departments are deleted by:-

```
DELETE FROM Department
WHERE ManagerID = 6381;
```

    a) All departments
    b) Sales Only
    c) Marketing only
    d) Sales and Marketing

14. What is missing to delete only **Technical support**?

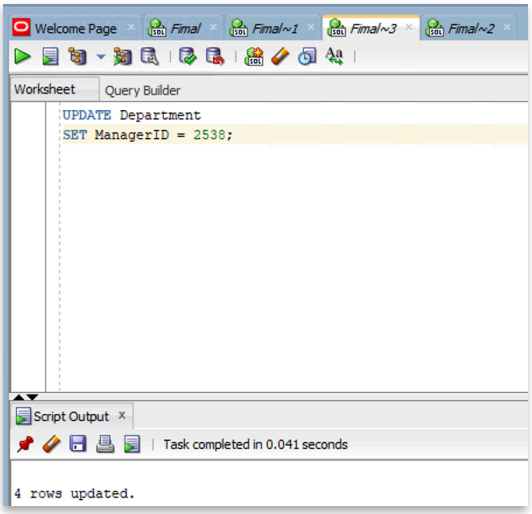
```
DELETE FROM Department
WHERE ___;
```

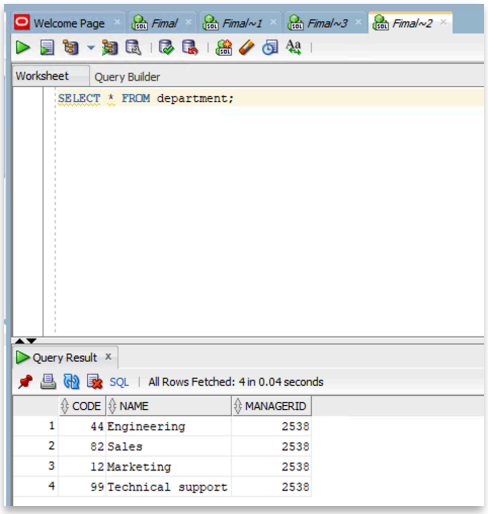
a) ManagerID = NULL
b) Name = “Technical Support”
c) Code = 99
d) Code = 99 and ManagerID = NULL

15. What department managers are changed?

```
UPDATE Department
SET ManagerID = 2538;
```

a) All managers
b) No managers
c) Only the Engineering Manager
d) Only the Technical Support Manager





Part B:- Answer **all 3** questions in spaces provided. Write all answers in spaces provided.

Question 1:- [5 Points]

You have been assigned to design the database for a new soccer club. Indicate the most appropriate sequence of activities by labeling each of the following steps in the correct order. (For example, if you think that "Load the database" is the appropriate first step, label it "1.")

__10__ Create the application programs.

___4__ Create a description of each system process.

__11__ Test the system.

___9__ Load the database

___7__ Normalize the conceptual model.

___2__ Interview the soccer club president.

___6__ Create a conceptual model using ER diagrams.

___3__ Interview the soccer club director of coaching.

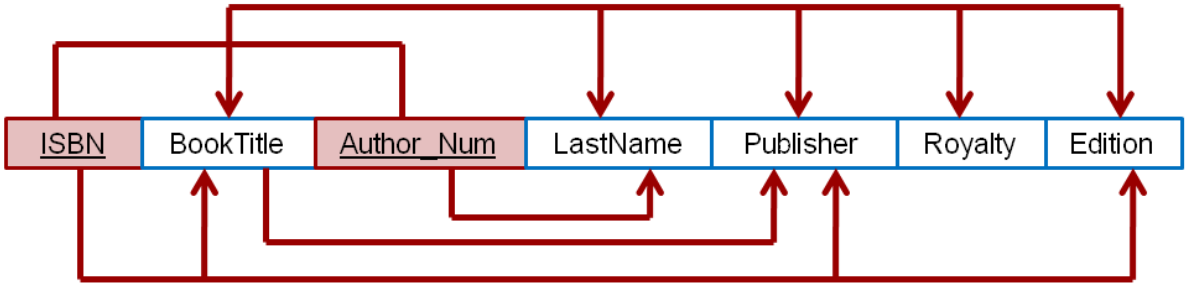
___8__ Create the file (table) structures.

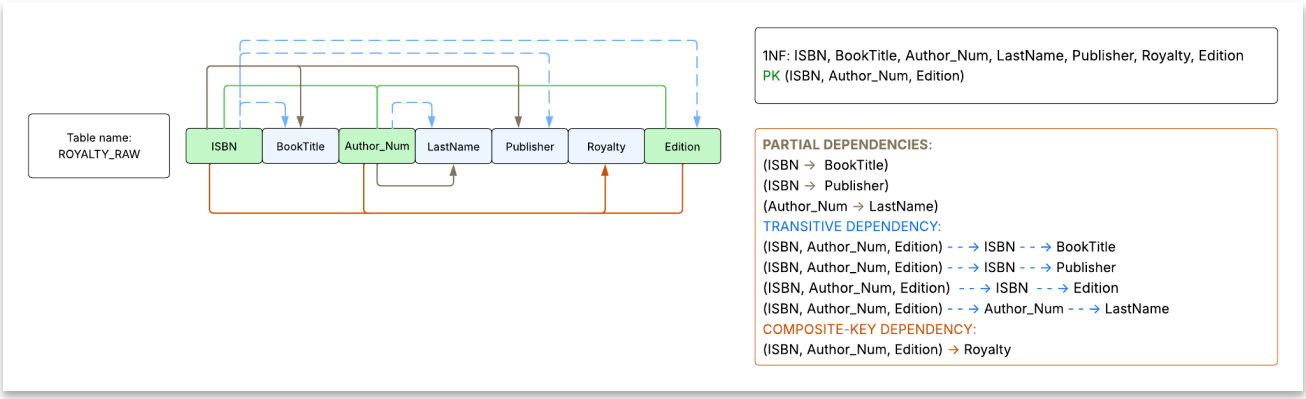
___1__ Obtain a general description of the soccer club operations.

___5__ Draw a data flow diagram and system flowcharts.

Question 2:- [30 Points]

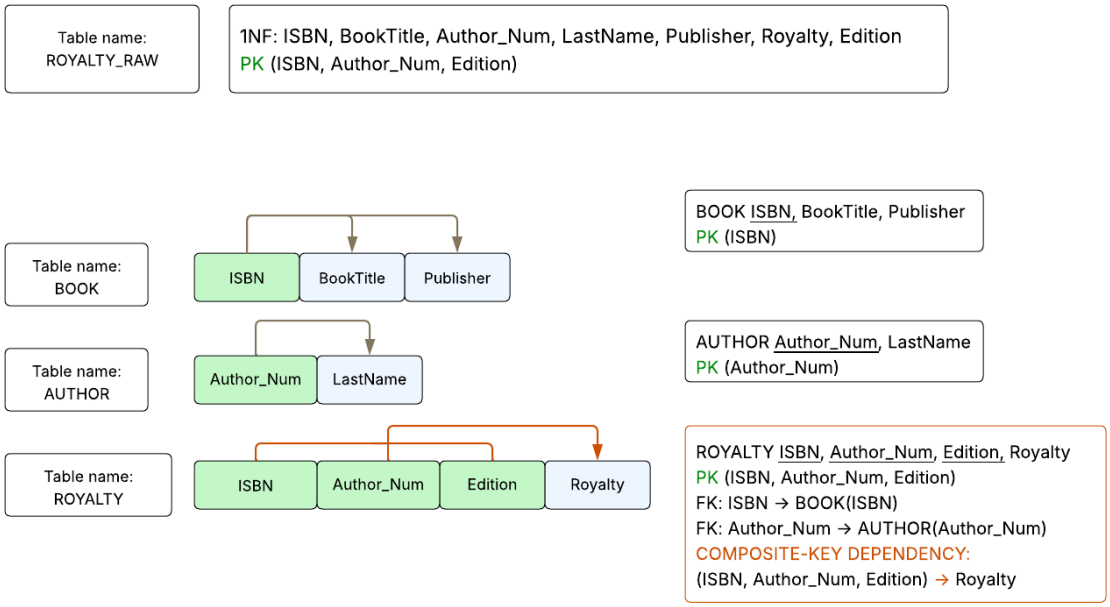
The dependency diagram shown indicates that authors are paid royalties for each book that they write for a publisher. The amount of the royalty can vary by author, by book, and by edition of the book.





a) **Based on the dependency diagram, create a database whose tables are at least in 2NF, showing the dependency diagram for each table. For full points, make sure to clearly label all partial and transitive dependencies.**

[10 Points]



Original relation (1NF)

ROYALTY_RAW(ISBN, BookTitle, Author_Num, LastName, Publisher, Royalty, Edition)

Primary key (PK): (ISBN, Author_Num, Edition)

Tables in at least 2NF (actually 3NF) and their dependency diagrams

1) BOOK table

- All non-key attributes depend on the whole key (ISBN).
- Partial dependencies: none.
- Transitive dependencies: none.
- Normal form: 3NF (therefore also 2NF).

2) AUTHOR table

- All non-key attributes depend on the whole key (Author_Num).
- Partial dependencies: none.
- Transitive dependencies: none.

- Normal form: 3NF (therefore also 2NF).

3) ROYALTY table

- Royalty depends on the entire composite key.
- Partial dependencies: none.
- Transitive dependencies: none.
- Normal form: 3NF (therefore also 2NF).

Based on the original dependency diagram, we identified and labeled all partial and transitive dependencies in ROYALTY_RAW, then decomposed the relation into BOOK, AUTHOR, and ROYALTY. Each resulting table has its own dependency diagram and is at least in Second Normal Form (in fact, all three tables are in Third Normal Form)

b) Remove all partial and transitive dependencies, write the relational schema, and draw the new dependency diagrams. Identify the normal forms for each table structure you created. Show all dependency diagrams and relational schema for 2NF and 3NF
[10Points]

After removing all partial and transitive dependencies from the original relation, the database decomposes into three tables.
Each table is shown below with its relational schema, dependency diagram, and normal form.

2NF Relational Schema
BOOK(ISBN, BookTitle, Publisher)
AUTHOR(Author_Num, LastName)
ROYALTY(ISBN, Author_Num, Edition, Royalty)

All partial dependencies have been eliminated.
Dependency Diagrams and Normal Forms

1. BOOK
Relational Schema
BOOK(ISBN, BookTitle, Publisher)
PK: ISBN

Dependency Diagram
ISBN → BookTitle
ISBN → Publisher

Normal Form:
No partial or transitive dependencies → 2NF (also 3NF)

2. AUTHOR
Relational Schema
AUTHOR(Author_Num, LastName)
PK: Author_Num
Dependency Diagram
Author_Num → LastName
Normal Form:
No partial or transitive dependencies → 2NF (also 3NF)

3. ROYALTY
Relational Schema
ROYALTY(ISBN, Author_Num, Edition, Royalty)
PK: (ISBN, Author_Num, Edition)
FK: ISBN → BOOK(ISBN)
FK: Author_Num → AUTHOR(Author_Num)

Dependency Diagram
(ISBN, Author_Num, Edition) → Royalty

Normal Form:
Royalty depends on the full composite key; no partial or transitive dependencies → 2NF (also 3NF)

3NF Relational Schema

Because none of the 2NF tables contain transitive dependencies, the 3NF schema is identical to the 2NF schema:

BOOK(ISBN, BookTitle, Publisher)
AUTHOR(Author_Num, LastName)
ROYALTY(ISBN, Author_Num, Edition, Royalty)

All partial and transitive dependencies were removed from the original relation.

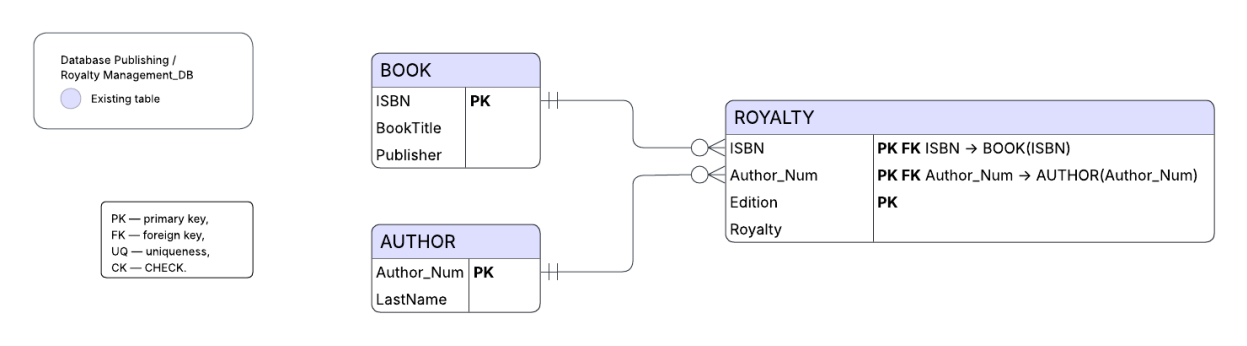
The resulting tables BOOK, AUTHOR, and ROYALTY are each in 2NF, and since none contain transitive dependencies, they also satisfy 3NF.

Note:

The 2NF and 3NF schemas are identical.

After removing all partial and transitive dependencies, the resulting tables (BOOK, AUTHOR, ROYALTY) already satisfy the conditions for 3NF, so no further decomposition is required.

c. Draw the Crow's Foot ERD. [10 Points]



Book–Royalty relationship

[BOOK] || —— o{ [ROYALTY]

(A book may appear in zero or many royalty records; each royalty record refers to exactly one book)

Author–Royalty relationship

[AUTHOR] || —— o{ [ROYALTY]

(An author may appear in zero or many royalty records; each royalty record refers to exactly one author)

Composite authorship (implicit via ROYALTY)

[BOOK] || —— o{ [ROYALTY] o} —— || [AUTHOR]

(ROYALTY is the associative entity representing the M:N relationship between BOOK and AUTHOR, distinguished further by Edition)

Edition as part of the identifying relationship

[ROYALTY] identifies each royalty entry by (ISBN, Author_Num, Edition)

Each book–author pair may have zero or many editions recorded in ROYALTY.

Interpretation:

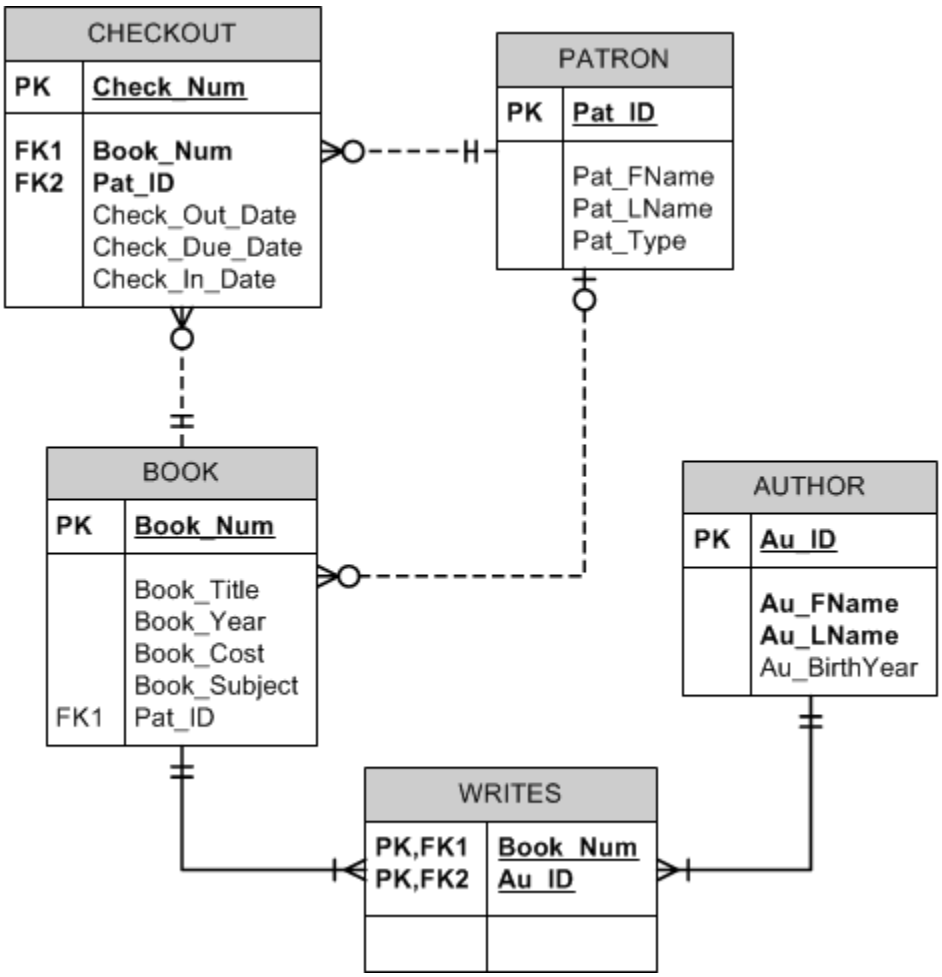
- A book may have zero or many royalty entries (for different authors or editions).
- An author may have zero or many royalty entries (for different books or editions).
- Each royalty record belongs to exactly one book and one author.

Question 3:- [20 Points]

Use the ERD diagram below to answer parts (a) to (f). Use the '**Final Exam Data File.txt**' provided under the Final Exam section in Canvas to create the tables and insert data.

Note:-

1. This is the **same dataset used for Problem Set 4** – if you already created the tables and inserted data, no need to recreate.
2. Note that the sample output may not include all rows from the query results.



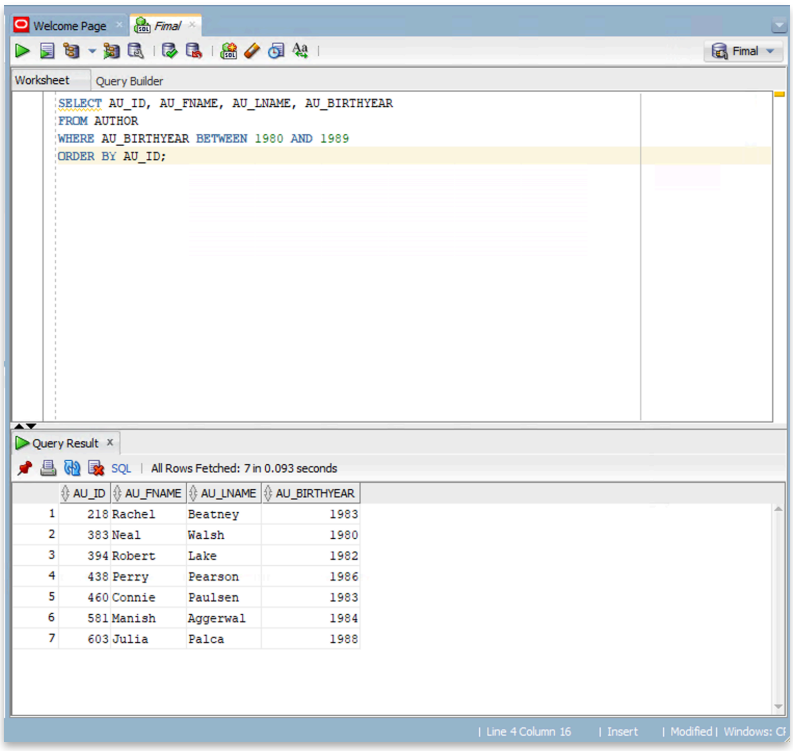
Name:- Siarhei Hancharou **Id#:-1712666**

a) Write a query to display the author ID, first name, last name, and year of birth for all authors born in the decade of the 1980s sorted by author ID [5 Points]

Sample Output:- Authors Born in the 1980's

AU_ID	AU_FNAME	AU_LNAME	AU_BIRTHYEAR
218	Rachel	Beatney	1983
383	Neal	Walsh	1980
394	Robert	Lake	1982
438	Perry	Pearson	1986
460	Connie	Paulsen	1983
581	Manish	Aggerwal	1984
603	Julia	Palca	1988

```
SELECT AU_ID, AU_FNAME, AU_LNAME, AU_BIRTHYEAR
FROM AUTHOR
WHERE AU_BIRTHYEAR BETWEEN 1980 AND 1989
ORDER BY AU_ID;
```



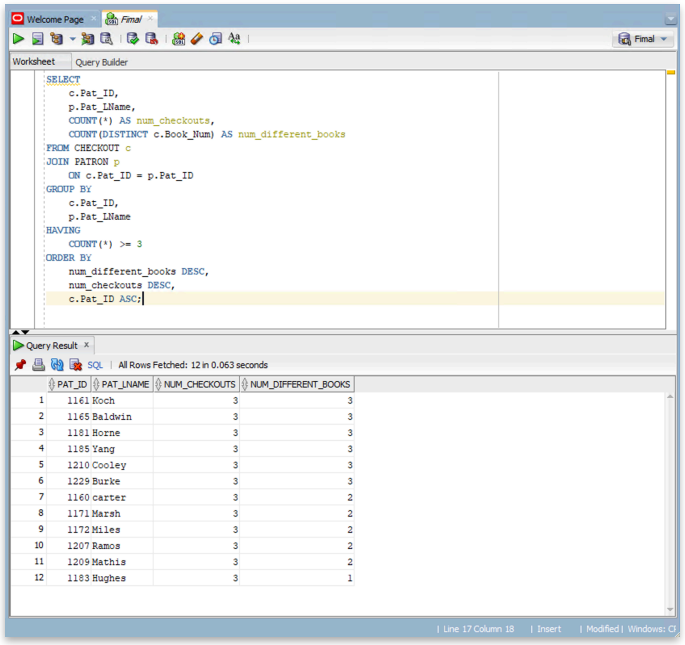
Name:- Siarhei Hancharou **Id#:-1712666**

b) Write a query to display the patron ID, last name, number of times that patron has ever checked out a book, and the number of different books the patron has ever checked out. For example, if a given patron has checked out the same book twice, that would count as two checkouts but only one book. Limit the results to only patrons who have made at least three checkouts. Sort the results in descending order by number of books, then in descending order by number of checkouts, and then in ascending order by patron ID.
[5 Points]

Sample Output:- Chekouts and Books by Patron

PAT_ID	PAT_LNAME	NUM CHECKOUTS	NUM DIFFERENT BOOKS
1161	Koch	3	3
1165	Baldwin	3	3
1181	Horne	3	3
1185	Yang	3	3
1210	Cooley	3	3
1229	Burke	3	3
1160	carter	3	2
1171	Marsh	3	2
1172	Miles	3	2
1207	Ramos	3	2
1209	Mathis	3	2
1183	Hughes	3	1

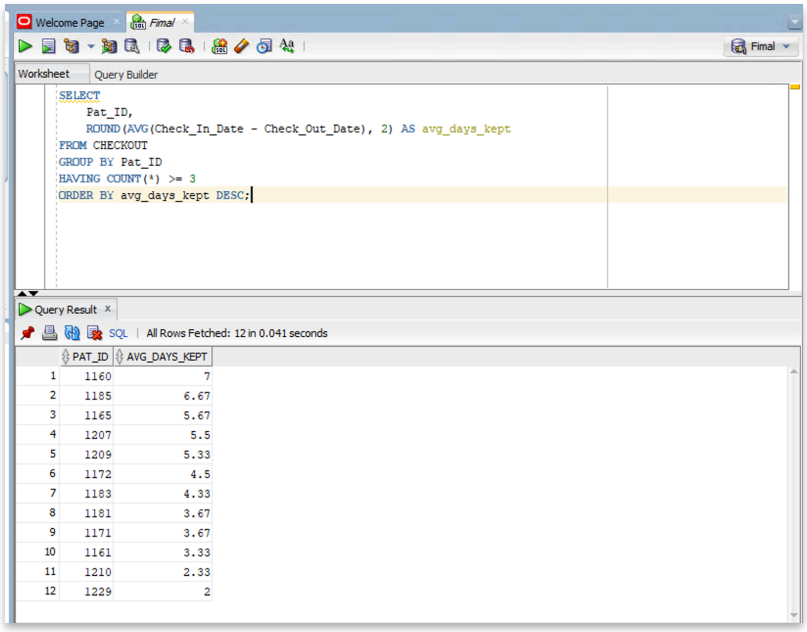
SELECT
```
   c.Pat_ID,
   p.Pat_LName,
   COUNT(*) AS num_checkouts,
   COUNT(DISTINCT c.Book_Num) AS num_different_books
FROM CHECKOUT c
JOIN PATRON p
   ON c.Pat_ID = p.Pat_ID
GROUP BY
   c.Pat_ID,
   p.Pat_LName
HAVING
   COUNT(*) >= 3
ORDER BY
   num_different_books DESC,
   num_checkouts DESC,
   c.Pat_ID ASC;
```



c) Write a query to display the patron ID and the average number of days that patron keeps books during a checkout. Limit the results to only patrons who have **at least three** checkouts. Sort the results in descending order by the average days the book is kept. [5 Points]

Sample Output:-Average days kept by Patron

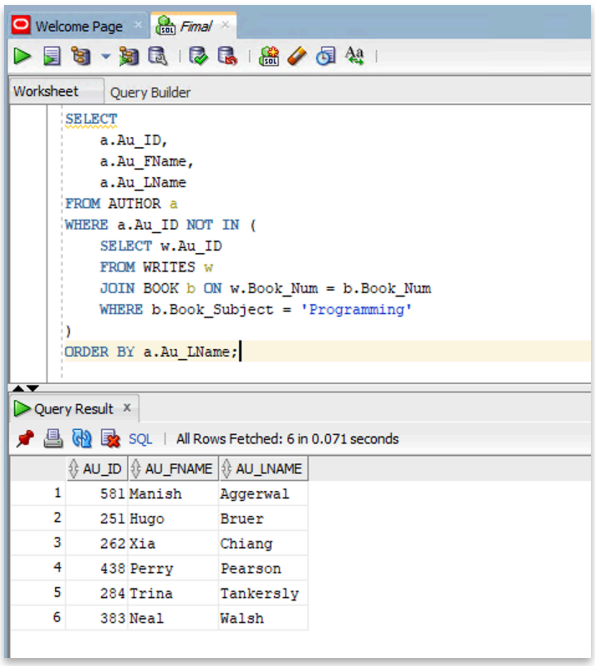
PAT_ID	Average Days Kept
1160	7
1185	6.67
1165	5.67
1207	5.5
1209	5.33
1172	4.5
1183	4.33
1181	3.67
1171	3.67
1161	3.33
1210	2.33
1229	2




```
SELECT
   Pat_ID,
   ROUND(AVG(Check_In_Date - Check_Out_Date), 2) AS avg_days_kept
FROM CHECKOUT
GROUP BY Pat_ID
HAVING COUNT(*) >= 3
ORDER BY avg_days_kept DESC;
```

d) Write a query to display the author ID, first and last name for all authors who have never written a book with the subject Programming. Sort the results by author last name. [5 Points]

AU_ID	AU_FNAME	AU_LNAME
581	Manish	Aggerwal
251	Hugo	Bruer
262	Xia	Chiang
438	Perry	Pearson
284	Trina	Tankersly
383	Neal	Walsh




```
SELECT
   a.Au_ID,
   a.Au_FName,
   a.Au_LName
FROM AUTHOR a
WHERE a.Au_ID NOT IN (
   SELECT w.Au_ID
   FROM WRITES w
   JOIN BOOK b ON w.Book_Num = b.Book_Num
   WHERE b.Book_Subject = 'Programming'
)
ORDER BY a.Au_LName;
```